1. Let $A = \{x \in \{0,1\}^* \mid x$ has an equal number of occurrences of the substrings 01 and 10$\}$. Show that A is regular. **(6 marks)**

2. Let A be the set of strings over $\{0,1\}$ such that every block of five consecutive symbols contains at least two 0's. Show that A is regular. **(8 marks)**

3. Let A be the set of all strings over $\{0,1\}$ with an equal number of 0's and 1's, such that each prefix has at most one more 0 than 1's and at most one more 1 than 0's. Show that A is regular. **(8 marks)**

4. Let A be the following set $\{xwx^R \mid x, w \in \{0,1\}^* \setminus \{\varepsilon\}$ and x^R is reverse of $x\}$. Is A regular or not? Justify your answer. **(8 marks)**

5. Let M_1 and M_2 be two DFAs given as input. Give a polynomial-time algorithm to decide whether the two automata accepts the same language, i.e., $L(M_1) = L(M_2)$. **(10 marks)**

6. Show that the language $L(M)$ recognized by a deterministic finite automaton M with k states is infinite if and only if it accepts a string of length ℓ where $k \leq \ell < 2k$. **(12 marks)**

7. For any $A \subseteq \Sigma^*$, define

$$A^{\mathsf{s}} := \{w_2w_1w_4w_3 \ldots w_{2n}w_{2n-1} \mid w = w_1w_2w_3w_4 \ldots w_{2n-1}w_{2n} \in A\}.$$

Show that if A is regular, then A^{s} is also regular. **(12 marks)**

8. For any $A \subseteq \Sigma^*$, define

$$A_p := \{x \mid xx^R \in A \text{ where } x^R \text{ is reverse of } x\}.$$

Is it true that if A is regular, then A_p is also regular? Justify your answer. **(12 marks)**

9. Show that the set $\{a^m b^n c^n \mid m, n \geq 1\}$ is not regular. **(12 marks)**

10. Show that the set $\{a^p \mid p$ is a prime$\}$ is not regular. **(12 marks)**